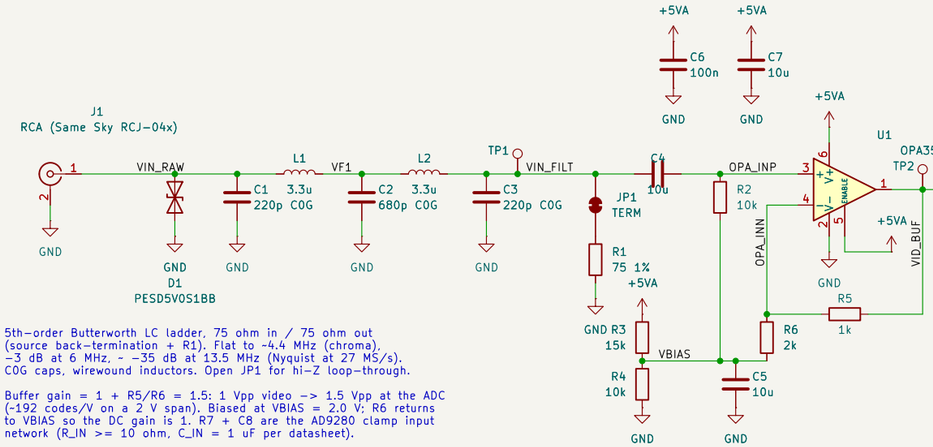
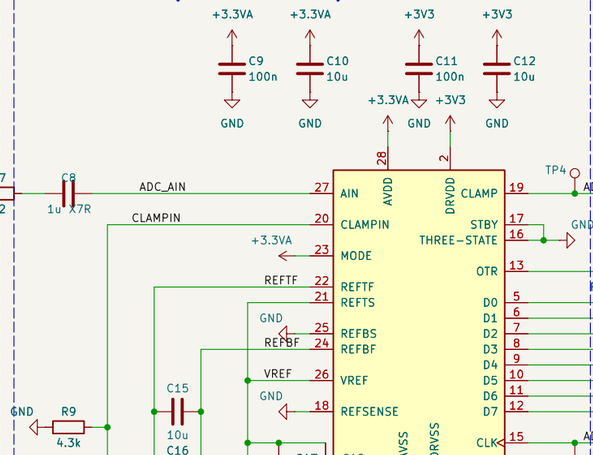
VIDEO INPUT, 6 MHz ANTI-ALIAS FILTER, BUFFER



5th-order Butterworth LC ladder, 75 ohm in / 75 ohm out (source back-termination + R1). Flat to -4.4 MHz (chroma), -3 dB at 6 MHz, - -35 dB at 13.5 MHz (Nyquist at 27 MS/s). C0G caps, wirewound inductors. Open JP1 for hi-Z loop-through.

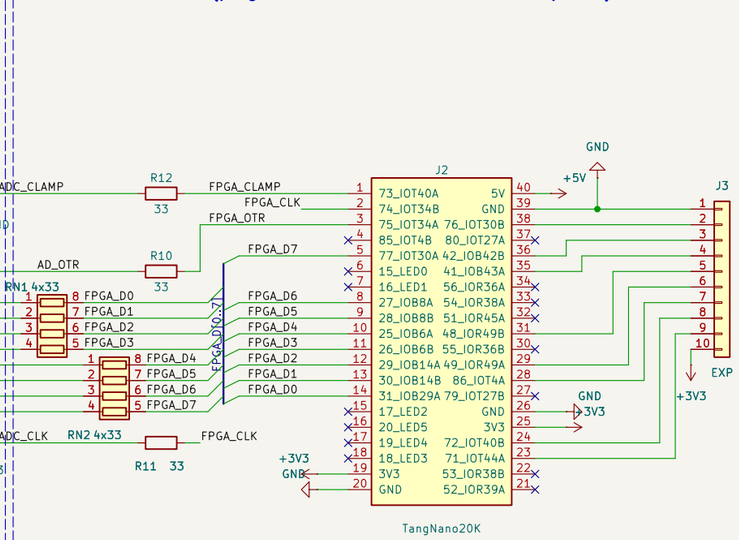
Buffer gain = 1 + R5/R6 = 1.5: 1 Vpp video -> 1.5 Vpp at the ADC (~192 codes/V on a 2 V span). Biased at VBIAS = 2.0 V; R6 returns to VBIAS so the DC gain is 1. R7 + C8 are the AD9280 clamp input network (R_IN >= 10 ohm, C_IN = 1 uF per datasheet).

AD9280 ADC (8 bit, 27 MS/s)



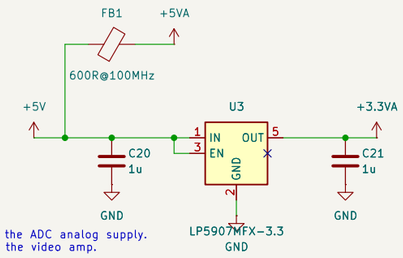
Top/bottom mode, 2 V span: MODE=AVDD, REFSENSE=GND, REFTS=VREF (2 V), REFBS=GND -> AIN 0..2 V. CLAMPIN = 2 V x 4.3k/14.3k = 0.60 V -> back porch at code ~77, sync tip ~19, white ~211. DRVDD from the Tang Nano 3V3: 3.3 V CMOS outputs.

TANG NANO 20K (plugs in, USB-C end = header pin 1)



33 ohm series R at the source end of each line.
Header pins 6-7 and 15-18 are the on-board LEDs: leave free.
FPGA 32-40 are not on the header (33-40 drive HDMI).

POWER (from the Tang Nano header: 5V = USB VBUS, 3V3 = on-board regulator)



+3.3VA: quiet LDO for the ADC analog supply.
+5VA: filtered 5 V for the video amp.



MECHANICAL



Pin map must match constraints/tangnano20k.cst
PAL composite in -> 6 MHz LC filter -> OPA355 buffer -> AD9280 -> Tang Nano 20K

Clacktronics

Sheet: /
File: tbc_carrier.kicad_sch

Title: Clacktronics TBC — Tang Nano 20K carrier

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 1/1